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User Stories - RabbitCTF

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FR-1: User management

- **FR-1.1: Register users:** The system must allow new users to register using a form that requests a username, email address, and password, for the purpose of creating unique and persistent accounts to access the platform and participate in the CTF competition. The system must validate that all required fields are complete and that the email address and username are unique before creating the account.

FR-1.1: Register users (5)**User Story**

As a new user

I need the system to allow me to register with a username, email, and password

So that I can create a unique account and access the platform.

Acceptance Criteria

Given that I am on the registration form

When I submit my username, email, and password

Then the system must validate that fields are complete and that the username and email are unique before creating the account.

- **FR-1.2: User login:** The system must allow users and administrators to log in by validating their credentials, consisting of a username and password, establishing authenticated access that enables views and permissions according to the assigned role. The session must remain active until the user manually closes it, the event ends, or the administrator revokes access.

FR-1.2: User Login (13)**User Story**

As a user or administrator

I need to log in with my username and password

So that I can access the system according to my role.

Acceptance Criteria

- ❖ **Given** that I have an existing account
When I provide valid login credentials
Then the system must authenticate me and keep my session active until logout, or admin revocation.
- ❖ **Given** that I attempt to log in
When I enter an incorrect username or password
Then the system must deny access and display an error message indicating that the credentials are invalid.

- **FR-1.3: Register initial system administrator:** The system must allow, only during the first run when there are no registered administrators, the creation of an initial administrator using a form that requests a username, email address, and password. Once the first administrator has been registered, the system must permanently disable this option to prevent new initial administrators from being registered in this way.

FR-1.3: Register initial system administrator (5)**User Story**

As a system operator on first run

I need to register the initial administrator

So that the system has its primary administrative account.

Acceptance Criteria

Given that no administrators exist in the system

When I create the first administrator account

Then the system must allow its creation and permanently disable this option afterward.

- **FR-1.4 – Create teams:** The system must allow an authenticated user who does not belong to any team to create a new team using a form that requests a name and password.

The system must validate that the team name is unique, automatically register the creator as captain, and enable the team so that other users can join using the defined password.

FR-1.4: Create teams (5)**User Story**

As an authenticated user without a team

I need to create a new team by providing a name and password

So that I can be assigned as the captain and enable others to join my team.

Acceptance Criteria

Given I do not belong to any team, and I am on the form to create a new team (enabled only for people who do not have a team)

When I submit a unique name and password on the form.

Then the system must create the team and assign the user who created it as captain.

- **FR-1.5 – Consult teams:** The system must allow authenticated users to consult a general list of registered teams, showing the name and current number of members for each team.

FR-1.5: Consult teams (3)**User Story**

As an authenticated user

I need to view a list of all registered teams

So that I can see their names and member counts.

Acceptance Criteria

Given that I am authenticated

When I access the teams list

Then the system must display each team's name and number of members and their current score.

- **FR-1.6 – Viewing information about one's own team:** The system must allow an authenticated user who belongs to a team to view detailed information about their team, including the captain's name, the complete list of members, and the total score accumulated in the competition.

FR-1.6: Viewing information about one's own team (3)**User Story**

As an user who belongs to a team

I need to view detailed information about my team

So that I can see the captain, members, and score.

Acceptance Criteria

Given that I am authenticated

When I view my team section

Then the system must show captain, members, and total score.

- **FR-1.7 – Joining teams:** The system must allow an authenticated user to join an existing team by entering the team name and access password. The system must validate that the user does not currently belong to any team, that the password provided is correct, and that the team has not reached the maximum number of members allowed before completing the addition.

FR-1.7: Joining teams (5)**User Story**

As an authenticated user who does not belong to a team

I need to join a team by entering its name and password

So that I can participate as part of that team.

Acceptance Criteria

Given that I am not currently on a team

When I enter the correct team name and password

Then the system must add me to the team if it is not full.

- **FR-1.8 – Deleting teams:** The system must allow a captain or administrator to delete a team, subject to explicit confirmation of the action. When deleting the team, the system must also delete all entries and scores associated with the team and its members.

FR-1.8: Deleting teams (5)

User Story

As a captain or administrator

I need to delete a team

So that the team and its associated scores are removed.

Acceptance Criteria

Given that I am a captain or administrator

When I confirm the team deletion

Then the system must remove the team and all related scores and entries.

- **FR-1.9 – Request password recovery:** The system must allow a user who has forgotten their password to request that the administrator reset their account using a form available on the login interface. The system must generate and record the request, notifying the user that their request has been sent to the administrator and keeping this notification visible until the user closes it manually.

FR-1.9: Request password recovery (13)

User Story

As a user who forgot their password

I need to request that an administrator reset my account
So that I can regain access.

Acceptance Criteria

Given that I am on the login interface

When I submit a password recovery request

Then the system must record the request and show a confirmation message.

- **FR-1.10 – Reset user password:** The system must allow the administrator to reset a user's password from the administration panel, either by setting a new password or generating a temporary password that allows the user to access their account again.

FR-1.10: Reset user password (8)

User Story

As an administrator

I need to reset a user's password

So that the user can access their account again.

Acceptance Criteria

Given that I am in the admin panel of users

When I assign or generate a new password

Then the user account must update with the new password or with a temporary one.

FR-2: Challenge management

- **FR-2.1 – Create challenges:** The administrator must be able to create a new challenge by entering its main information, including title, description, category, tags, difficulty level, score, scoring mode, and operational data such as URL or port when applicable, leaving the challenge initially in draft status or ready for publication according to the established configuration.

FR-2.1: Create a new challenge (8)**User Story**

As an administrator

I need to create a new challenge by entering all required information

So that I can prepare it for publication or leave it in draft

Acceptance Criteria

Given I am an authenticated administrator creating a challenge

When I enter the title, description, category, tags, difficulty, score, scoring mode, and URL/port if applicable

Then the system must create the challenge and set its status to draft or ready for publication based on configuration.

- **FR-2.2 – Configure challenge visibility:** The administrator must be able to define the visibility of the challenge at the time of its creation or editing, establishing whether it will be publicly available to competitors or remain in draft status until its manual publication.

FR-2.2: Configure challenge visibility (3)**User Story**

As an administrator

I need to set whether a challenge is public or remains in draft

So that I can control when competitors can access it

Acceptance Criteria

Given I am creating or editing a challenge

When I configure its visibility

Then the system must apply that visibility, restricting or granting competitor access accordingly.

- **FR-2.3 – Edit existing challenges:** The administrator must be able to modify any attribute of an existing challenge without affecting the history of flag submissions or previously recorded score references.

FR-2.3: Edit challenge information (5)**User Story**

As an administrator

I need to modify any attribute of an existing challenge

So that I can correct or improve challenge information without altering submission history

Acceptance Criteria

Given a challenge already exists

When I update any of its attributes

Then the system must save the changes without affecting flag submission history or recorded scoring data.

- **FR-2.4 – Manage flag and sensitivity:** The administrator must be able to set the flag associated with a challenge and define whether its validation will be case-sensitive or not.

FR-2.4: Configure flag and case sensitivity (3)**User Story**

As an administrator

I need to set the challenge flag and its case-sensitivity mode

So that I can enforce correct validation behavior

Acceptance Criteria

Given I am configuring a challenge

When I define the flag and select case sensitivity or insensitivity

Then the system must validate competitor submissions according to the selected mode.

- **FR-2.5 – Configure scoring mode:** The administrator must be able to assign the scoring mode for the challenge, choosing between a static or dynamic value, and in the latter case, define the decay factor and the minimum achievable score.

FR-2.5: Configure scoring mode (3)
User Story As an administrator I need to choose between static and dynamic scoring for a challenge So that I can define how points are awarded during the event
Acceptance Criteria Given I am editing the scoring settings When I select static or dynamic scoring and configure decay factor and minimum score (for dynamic) Then the system must apply the selected scoring mode and parameters.

- **FR-2.6 – Attach files to the challenge:** The administrator must be able to attach one or more files to the challenge during its creation or editing, ensuring that the files are linked to the challenge and available to competitors according to the configured visibility.

FR-2.6: Attach files to a challenge (5)
User Story As an administrator I need to upload one or more files associated with a challenge So that I can provide resources needed by competitors
Acceptance Criteria Given I am creating or editing a challenge When I upload files

Then the system must attach them to the challenge and make them available to competitors according to visibility rules.

- **FR-2.7 – Delete challenges with explicit confirmation:** The administrator must be able to delete a challenge after explicitly confirming the action, ensuring that the deletion is only executed after such confirmation.

FR-2.7: Delete challenge with confirmation (1)

User Story

As an administrator

I need to confirm before deleting a challenge

So that I can prevent accidental loss of data

Acceptance Criteria

Given I select a challenge for deletion

When I explicitly confirm the action

Then the system must permanently delete the challenge; if I cancel, no deletion must occur.

- **FR-2.8 – Ensure consistency after challenge deletion:** The system must delete all attached resources and accessible references associated with the challenge, maintaining consistency with historical records in accordance with the established retention policy.

FR-2.8: Maintain consistency after challenge deletion (8)

User Story

As an administrator

I need to ensure all associated resources of a deleted challenge are removed

So that I can keep the system consistent with retention policies

Acceptance Criteria

Given a challenge has been deleted

When the deletion is executed

Then the system must remove all attachments and references while keeping historical records consistent with policy.

- **FR-2.9 – Attach files to challenges:** The administrator must be able to attach one or more files to a challenge during its creation or editing, respecting a maximum individual size of 100 MB and a cumulative limit of 500 MB per challenge.

FR-2.9: File size restrictions for challenge attachments (5)

User Story

As an administrator

I need to upload files within allowed size limits

So that I can manage storage usage properly

Acceptance Criteria

Given I am uploading files to a challenge

When an individual file exceeds 100 MB or total attachments exceed 500 MB

Then the system must reject the upload and notify the administrator.

- **FR-2.10 – Delete files associated with the challenge:** The administrator must be able to delete files previously attached to a challenge, ensuring that they are removed from both storage and system references.

FR-2.10: Delete challenge files (3)

User Story

As an administrator

I need to delete files attached to a challenge

So that I can remove outdated or incorrect files from storage and system references

Acceptance Criteria

Given a challenge has one or more attached files

When I remove a file through the challenge management interface

Then the system must delete the file from storage and remove all references from the challenge.

- **FR-2.11 – Create challenge categories:** The administrator must be able to create new challenge categories to facilitate their classification and filtering within the system.

FR-2.11: Create challenge categories (5)

User Story

As an administrator

I need to create new categories

So that I can classify and filter challenges more effectively

Acceptance Criteria

Given I access the category management view

When I create a new category with a name and optional description

Then the system must store the category and make it available for challenge classification.

- **FR-2.12 – Edit challenge categories:** The administrator must be able to modify the name or description of an existing category without affecting the association of challenges already classified under it.

FR-2.12: Edit challenge categories (5)

User Story

As an administrator

I need to update the name or description of a category
So that I can keep category information accurate and relevant

Acceptance Criteria

Given a category already exists

When I modify its name or description

Then the system must save the changes without affecting the challenges already assigned to that category.

- **FR-2.13 – Deactivate challenge categories:** The administrator must be able to deactivate a challenge category, preventing it from being used in new creations or edits, without removing it from the system or affecting previously classified challenges.

FR-2.13: Deactivate challenge categories (5)

User Story

As an administrator

I need to deactivate a category

So that I can prevent its use in new challenge creation or editing

Acceptance Criteria

Given a category exists

When I deactivate it

Then the system must prevent the category from being used for new or edited challenges

And must keep the category available for previously classified challenges.

- **FR-2.14 – Delete challenge categories:** The administrator must be able to delete a challenge category only when there are no active challenges associated with it, ensuring the integrity of references.

FR-2.14: Delete inactive challenge categories (3)**User Story**

As an administrator

I need to delete a category only when it has no active challenges

So that I can maintain clean and consistent category organization

Acceptance Criteria

Given a category exists in the system

When I attempt to delete it

Then the system must allow deletion only if no active challenges are associated with it

And must prevent deletion otherwise.

- **FR-2.15 – Display challenge solution history:** The system must display a solution history on the challenge modal that includes, at a minimum, the user or team that solved it and the date and time of the solution.

FR-2.15: Display challenge solution history (5)**User Story**

As a system

I need to display a solution history for each challenge

So that I can inform users about who solved it and when

Acceptance Criteria

Given a challenge has recorded solutions

When a user opens the challenge modal

Then the system must display, at minimum, the user or team who solved it and the exact date and time of the solution.

FR-3: Participation and validation

- **FR-3.1 – Accessing the challenge panel:** The authenticated competitor must be able to access a panel showing all available challenges to which their team has access.

FR-3.1: Access and display challenge panel (5)**User Story**

As a competitor

I need to access a panel listing all available challenges

So that I can view and attempt the challenges assigned to my team

Acceptance Criteria

Given I am an authenticated competitor

When I navigate to the challenge dashboard

Then the system must display all challenges available to my team.

- **FR-3.2 – Display key information for each challenge:** For each available challenge, the system must display its title, category, difficulty level, current score, and current status with respect to the user's team (solved or unsolved) on the dashboard.

FR-3.2: Display essential challenge information (3)**User Story**

As a competitor

I need to see key information for each challenge

So that I can evaluate difficulty and choose which challenges to attempt

Acceptance Criteria

Given the challenge panel is displayed

When I view the list of available challenges

Then the system must show, for each challenge:

- title
- category
- difficulty level
- current score
- solved/unsolved status for my team

- **FR-3.3 – Submit challenge flag:** The competitor must be able to submit a text string as a flag for a published challenge using the corresponding form. The system must validate the flag and return the validation result within a maximum of 2 seconds without exposing sensitive information.

FR-3.3: Submit challenge flag (5)

User history

As a competitor

I need to submit a text string as a flag for a published challenge through its corresponding form

So that I can validate whether my solution to the challenge is correct and obtain immediate feedback.

Acceptance criteria

Given that I am a competitor and the challenge is published

When I submit a text string as a flag through the challenge's submission form

Then the system must validate the flag and return the result (correct or incorrect) within a maximum of 2 seconds.

Given that flag validation must be secure

When I receive the validation result

Then the system must not expose any sensitive information, internal logic, or hints about the correct flag.

- **FR-3.4 – Confirm correct flag:** When the flag sent matches the configured solution, the system must display a non-blocking confirmation message on the corresponding challenge interface, indicating success and disappearing automatically within a maximum of 8 seconds. The system must mark the challenge as solved for the corresponding team and update the team and user scores on the interface within a maximum of 5 seconds.

FR-3.4: Confirm correct flag (5)

User history

As a competitor

I need the system to confirm when the flag I submitted is correct

So that I can immediately know I solved the challenge and see my team and personal scores updated accordingly.

Acceptance criteria

Given that I submitted a flag and it matches the configured solution

When the system processes the submission

Then it must display a non-blocking confirmation message on the challenge interface indicating success, and this message must disappear automatically within a maximum of 8 seconds.

Given that the flag submission is correct

When the system records the result

Then it must mark the challenge as solved for my team and update both the team score and my user score on the interface within a maximum of 5 seconds.

- **FR-3.5 – Notify incorrect flag:** When the flag sent does not match the configured solution, the system must display a generic non-blocking message on the corresponding challenge interface, indicating an error without revealing details or patterns of the solution. The message must remain visible for 3 seconds before disappearing automatically.

FR-3.5: Notify incorrect flag (3)
User history As a competitor I need the system to notify me when the flag I submitted is incorrect So that I understand my attempt failed without receiving any sensitive hints or information about the correct solution.
Acceptance criteria Given that I submitted a flag and it does not match the configured solution When the system processes the submission Then it must display a generic non-blocking error message on the challenge interface, without revealing any clues, patterns, or details about the solution. Given that the message is non-blocking When it is shown Then it must remain visible for 3 seconds and disappear automatically afterward.

- **FR-3.6 – Log validation attempts:** The system must log each flag submission attempt, including user or team, challenge, date and time in UTC format, and validation result.

FR-3.6: Log validation attempts (5)
User history As a system administrator I need every flag submission attempt to be recorded So that I can audit challenge activity, monitor competitor behavior, and maintain a complete and secure history of validation events.
Acceptance criteria Given that a competitor submits a flag for any challenge When the system processes the submission Then it must record a log entry including the user or team, the challenge, the date and time in UTC format, and the validation result.

- **FR-3.7 – Apply validation settings:** The system must apply flag validation according to the configuration established for each challenge, considering case sensitivity defined for that challenge.

FR-3.7: Apply validation settings (3)
User history As a competitor I need the system to validate my submitted flag according to each challenge's specific configuration So that the evaluation is accurate, and respects case sensitivity rules.
Acceptance criteria Given that a challenge has defined validation settings (e.g., case sensitivity). When I submit a flag

Then the system must evaluate the flag strictly according to those settings.

- **FR-3.8 – View validation history:** The administrator must be able to view the complete validation history from the administrative panel, with filtering options by user, team, challenge, and date range. The system must return results in a maximum of 3 seconds for queries of up to 10,000 records.

FR-3.8: View validation history (5)

User history

As an administrator

I need to view the complete history of validation attempts with flexible filtering options

So that I can audit activity, detect anomalies, and analyze the performance and behavior of competitors across the event.

Acceptance criteria

Given that I am an authenticated administrator and there are recorded validation attempts

When I access the administrative panel and open the validation history section

Then I must be able to filter records by user, team, challenge, and date range.

Given a query containing up to 10,000 log records

When I apply any supported filter or perform a search

Then the system must return the filtered results within a maximum of 3 seconds.

- **FR-3.9 – Display validation statistics:** The administrator must be able to view aggregate statistics in the administrative panel, including the success rate per challenge, the average number of attempts before resolution, and the list of challenges with the highest number

of failed attempts. The system must calculate and display the updated values within a maximum of 10 seconds after the request.

FR-3.9: Display validation statistics (8)**User history**

As an administrator

I need to view aggregated validation statistics in the administrative panel

So that I can analyze challenge difficulty, identify problematic challenges, and understand overall competitor performance throughout the event.

Acceptance criteria

Given that validation attempts exist across multiple challenges

When I access the statistics section in the administrative panel

Then I must see aggregated metrics, including the success rate per challenge, the average number of attempts before resolution, and the list of challenges with the highest number of failed attempts.

Given that the administrator requests statistical data

When the system processes the request

Then it must calculate and display the updated statistics within a maximum of 10 seconds.

- **FR-3.10 – Apply score in static mode:** The system must assign exactly the score value configured for the challenge when it is in static mode, reflecting the result in the team's score and, when applicable, the user's score, within a maximum of 5 seconds after validation.

FR-3.10: Apply score in static mode (2)**User history**

As a competitor

I need the system to assign the exact configured score for a challenge when it uses static

scoring

So that my team's score and, when applicable, my personal score, accurately reflect the defined value without delay

Acceptance criteria

Given that a challenge is configured to use static scoring

When my submitted flag is validated as correct

Then the system must assign exactly the score value defined for that challenge.

Given that the score must be reflected promptly

When the score is assigned

Then the system must update the team's score and, when applicable, my user score within a maximum of 5 seconds after validation.

- **FR-3.11 – Apply scoring in dynamic mode:** The system must calculate the score in dynamic mode using the decay factor and minimum score configured, based on the number of solutions registered at the time of validation, and reflect the resulting score on the team and, when applicable, on the user, within a maximum of 5 seconds after validation.

FR-3.11: Apply scoring in dynamic mode (5)

User history

As a competitor

I need the system to calculate my score dynamically when I solve a challenge configured with dynamic scoring

So that the awarded points accurately reflect the difficulty and the decay rules based on how many competitors solved it before me.

Acceptance criteria

Given that a challenge is configured to use dynamic scoring

When my submitted flag is validated as correct

Then the system must calculate the score using the decay factor and minimum score

defined for that challenge, based on the number of registered solutions at the time of validation.

Given that the dynamically computed score must be applied promptly

When the score is calculated

Then the system must update the team's score and, when applicable, my personal score within a maximum of 5 seconds after validation.

- **FR-3.12 – Mark challenge as resolved for the team:** The system must automatically mark a challenge as “resolved” for all team members when any member has correctly validated the corresponding flag, ensuring that the status is visible to the entire team within a maximum of 5 seconds after validation.

FR-3.12: Mark challenge as resolved for the team (3)

User history

As a team member

I need the system to mark a challenge as resolved for my entire team when any teammate submits a correct flag

So that all members can immediately see the updated challenge status and avoid duplicate work.

Acceptance criteria

Given that any member of a team correctly validates a challenge's flag

When the system processes the successful validation

Then it must automatically mark the challenge as “resolved” for the entire team.

Given that the updated status must be visible to all members

When the challenge is marked as resolved

Then the system must reflect this status to every team member within a maximum of 5 seconds after validation.

- **FR-3.13 – Maintain access to resolved challenge:** The system must allow all team members, even when a challenge is in “resolved” status, to access its description, download associated files, and copy the challenge's operational data without restrictions.

FR-3.13: Maintain access to resolved challenge (3)**User history**

As a team member

I need to continue accessing a challenge even after it has been marked as resolved

So that I can review its description, download its files, and reuse any operational data without restrictions.

Acceptance criteria

Given that a challenge has been marked as “resolved” for my team

When I navigate to the challenge

Then the system must still allow me to view the full description, download any associated files, and copy the operational data.

Given that all team members may need reference material after solving a challenge

When the challenge is already resolved

Then no access restrictions must be applied and all informational elements must remain available.

- **FR-3.14 – Allow copying of visible challenge operational data:** The system must allow authenticated competitors to copy visible challenge operational data, such as addresses or short commands, to the clipboard, ensuring that the copied content corresponds exactly to the displayed text and does not include flags or sensitive information.

FR-3.14: Allow copying of visible challenge operational data (1)**User history**

As an authenticated competitor

I need to copy the visible operational data of a challenge (such as addresses or short commands)

So that I can quickly use this information during my attempt to solve the challenge without risking copying hidden or sensitive data.

Acceptance criteria

Given that I am an authenticated competitor and the challenge displays operational data

When I use the option to copy the data to the clipboard

Then the system must copy exactly the text that is visibly shown on the interface.

Given that sensitive information must remain protected

When the system performs the copy action

Then it must ensure that no hidden flags, solutions, or sensitive data are included in the copied content, only the visible operational text.

- **FR-3.15 – Visually confirm the copy action:** The system must display a visible confirmation to the user (competitor or administrator) immediately after executing the copy action, remaining visible for at least 2 seconds without blocking interaction with the platform.

FR-3.15: Visually confirm the copy action (1)

User history

As a competitor or administrator

I need the system to show a visible confirmation message when I copy operational data

So that I know the action was successful without interrupting my workflow or blocking my interaction with the platform.

Acceptance criteria

Given that I perform a copy action on visible operational data

When the system executes the copy operation

Then it must display a visible confirmation message immediately.

Given that the confirmation must be non-blocking

When the message is displayed

Then it must remain visible for at least 2 seconds and must not prevent me from continuing

to interact with the platform.

- **FR-3.16 – Count attempts and apply temporary blocking:** The system must count flag submissions per user or team when the limitation policy is active, and automatically block new submissions when the configured threshold is reached, restoring permission once the blocking time has expired.

FR-3.16: Count attempts and apply temporary blocking (13)

User history

As a competitor

I need the system to track my flag submission attempts and temporarily block me when I exceed the allowed threshold

So that brute-force attempts are prevented and the competition remains fair and secure.

Acceptance criteria

Given that a flag submission limitation policy is active

When I submit a flag

Then the system must count the attempt under my user or team according to the configured rules.

Given that the number of attempts reaches the configured threshold

When I try to submit an additional flag

Then the system must automatically block further submissions for the duration defined by the policy.

Given that the blocking time expires

When I attempt to submit a new flag

Then the system must restore my ability to submit flags normally, resetting or updating the attempt counter according to the policy rules.

- **FR-3.17 – Logging of blocking events due to submission limits:** The system must log each blocking event due to the submission limitation policy, storing only the timestamp, the user or team, and the challenge affected.

FR-3.17: Logging of blocking events due to submission limits (8)**User history**

As an administrator

I need the system to log every event in which a competitor or team is blocked due to exceeding the submission limit

So that I can audit platform behavior and detect potential abuse without exposing any sensitive information.

Acceptance criteria

Given that a competitor or team exceeds the configured submission limit

When the system enforces a temporary block

Then the system must create a log entry capturing only the timestamp, the user or team identifier, and the challenge affected.

Given that logs must not expose sensitive data

When the event is stored

Then the system must avoid saving any flag content or other sensitive information in plain text or indirectly.

Given that administrators may later review these events

When an audit is performed

Then the logged information must be sufficient to trace blocking events without compromising confidentiality.

- **FR-3.18 – Filter challenges by category:** The authenticated competitor must be able to filter the challenges visible on the panel according to the selected category, so that only challenges belonging to that category are displayed.

FR-3.18: Filter challenges by category (5)

User history

As a competitor

I need the system to show a generic error message when my submitted flag is invalid
So that I understand my attempt failed without receiving any clues or sensitive information about the correct solution.

Acceptance criteria

Given that I am authenticated and viewing the challenges panel

When I select a category from the available list

Then the system must display only the challenges that belong to the selected category.

Given that categories may change dynamically during the event

When I modify or clear the selected category filter

Then the system must immediately update the list of challenges to match the current filtering criteria, including showing all visible challenges when the filter is cleared.

Given that filtering must not reveal hidden challenges

When the filter is applied

Then the system must show only the challenges allowed by the event visibility policy, regardless of their category.

- **FR-3.19 – Filter challenges by difficulty level:** The authenticated competitor must be able to filter the challenges visible on the panel according to the selected difficulty level, displaying only the challenges that correspond to the chosen level.

FR-3.19: Filter challenges by difficulty level (3)**User history**

As a competitor

I need to filter the visible challenges by their difficulty level
So that I can focus on challenges appropriate to my skill level or intended progression.

Acceptance criteria

Given that I am authenticated and on the challenges panel

When I select a difficulty level from the available options
Then the system must display only the challenges that match the selected difficulty level.
Given that I may want to change or clear the filter
When I modify or remove the selected difficulty level
Then the system must immediately update the list to reflect the new filter, including showing all allowed challenges when no difficulty is selected.
Given that the event policy may restrict visibility of certain challenges
When I apply a difficulty filter
Then the system must only display challenges permitted by the visibility policy, even if some challenges match the selected difficulty.

- **FR-3.20 – Filter challenges by resolution status:** The authenticated competitor must be able to filter the challenges visible on the panel according to their status with respect to the team (resolved or unresolved), allowing them to focus on pending or completed challenges.

FR-3.20: Filter challenges by resolution status (3)

User history

As a competitor

I need to filter the visible challenges based on whether they are resolved or unresolved for my team

So that I can quickly focus on the challenges that are still pending or review those that have already been completed.

Acceptance criteria

Given that I am authenticated and viewing the challenges panel

When I select a filter for *resolved* or *unresolved* challenges

Then the system must display only the challenges matching the selected resolution status for my team.

Given that I may want to adjust or remove the status filter

When I change the filter selection or clear it

Then the system must immediately refresh the list to show the updated set of challenges, including all visible ones if no filter is applied.

Given that challenge visibility may also depend on event policy

When I apply the resolution status filter

Then the system must only display challenges permitted by the visibility rules, even if their status matches the filter.

FR-4: Rankings and statistics

- **FR-4.1 – View overall leaderboard:** The system must allow participants and administrators to view an event leaderboard showing teams with their accumulated scores, number of challenges solved, and sorted from highest to lowest score.

FR-4.1: View overall leaderboard (8)

User history

As a participant or administrator

I need to view the event's general leaderboard

So that I can know each team's position and the global progress of the CTF competition.

Acceptance criteria

- **Given that** there are registered teams and calculated scores
When I access the leaderboard view
Then I must be able to see the teams listed by score and number of solved challenges, sorted from highest to lowest.
- **Given that** there may be ties
When I access the leaderboard view
Then I must clearly notice a tie indication on the leaderboard.

- **FR-4.2 – View Participant and Team Counts:** The system shall provide administrators with a view of the total number of registered teams and the total number of registered participants.

FR-4.2: View participant and team registration totals (3)
User history As an administrator I need to view the total number of registered teams and participants So that I can track the scale and growth of the event's registration.
Acceptance criteria ● Given that teams and participants have registered for the event When I access the statistics panel and view the registration totals Then I must see the aggregate counts of total teams and total participants.

- **FR-4.3 – View Challenge Completion Statistics by Category:** The system shall provide administrators with a view of challenge completion statistics, displaying the number and percentage of challenges solved, broken down by challenge category (e.g., Web, Crypto, Forensics). This view must allow these statistics to be filtered by challenge category.

FR-4.3: View Challenge Completion Statistics by Category (8)
User history As an administrator I need to view the number and percentage of challenges solved, broken down by category So that I can identify which categories are most engaging and where participants are struggling.
Acceptance criteria

- **Given that** challenges have been attempted and solved
When I access the statistics panel and view the challenge completion section
Then I must see a list of all challenge categories with the count and percentage of solved challenges for each.

- **FR-4.4 – View Event Completion Percentage as a Team:** The system shall provide as a member of a team see an overall percentage of challenges completed out of the total available.

FR-4.4: View Event Completion Percentage as a Team (5)

User history

As a member of a team

I need to view the global percentage of challenges completed by my team

So that I can gauge the overall progress of my team in the event.

Acceptance criteria

- **Given that** the event is active and challenges are being solved
When I access the statistics panel
Then I must see a prominent indicator for my team showing the overall percentage of challenges completed against the total available.

- **FR-4.5 – View basic administrative statistics for the event:** The system must allow administrators to view a dashboard with event indicators that shows, in real time, the first to solve each challenge (first blood) and the outstanding player (MVP) calculated according to the highest individual score accumulated and the number of challenges solved.

FR-4.5: View basic administrative statistics for the event (5)

User history

As an administrator

I need to view the event performance indicators

So that I can recognize the performance of the top teams and players.

Acceptance criteria

- **Given that** teams are solving challenges
When I access the indicators view
Then I must see a list of first bloods and the MVP calculated automatically according to the defined criteria.
- **Given that** the policy may hide these indicators
When I access the indicators view
Then I must not be able to view the indicators in the interface.

FR-5: Communications and notifications

- **FR-5.1 – Display basic internal notifications on the platform:** The system must display relevant CTF events in an internal notifications section (e.g., publication of new challenges, announcements from the organizer).

FR-5.1: Display basic internal notifications on the platform (13)**User history**

As a participant or administrator

I need to receive notifications within the platform

So that I can be aware of important events without reloading the page.

Acceptance criteria

- **Given that** relevant events occur in the system
When I access the notifications section
Then I must see informative messages with type, description, and timestamp.
- **Given that** there may be multiple notifications

When I access the notifications interface

Then I can filter the notifications by type, displaying a brief description and timestamp per notification.

- **Given that** they can be received at any time

When a notification is received

Then the interface must display it in less than 5 seconds without fully reloading the page.

- **FR-5.2 – Optional integration with Discord via local configuration:** The system must allow the integration of a Discord webhook to publish basic CTF announcements (e.g., new challenges and featured captures).

FR-5.2: Optional integration with Discord via local configuration (20)

User history

As an administrator

I need to connect the platform to a Discord channel

So that I can post automatic announcements about new challenges or important captures.

Acceptance criteria

- **Given that** I have configured a valid webhook

When a relevant event is published

Then the system must send an informational message to Discord and stop sending if the integration is deactivated.

FR-6: Event configuration

- **FR-6.1 – Define CTF start and end date/time:** The system must allow administrators to define the start and end date and time of the CTF event, reflecting the CTF statuses in the interface.

FR-6.1 – Define CTF start and end date/time (5)**User history**

As an administrator

I need to define the event's start and end dates

So that I can control when participants can solve challenges and submit flags.

Acceptance criteria

- **Given that** the event has not yet started
When I define its start and end
Then the interface must display the event status and restrict or enable actions as appropriate, indicating the proximity of the CTF's start or end.
- **Given that** I am an authenticated administrator and have access to the event configuration view,
When I set or update the CTF's start and end date and time,
Then the system must automatically reflect the event status as not started, active, or finished, based on the current date and time.
- **Given that** dates have been defined for the event,
When a participant attempts to submit a flag,
Then the system must behave according to the event status:
 - ❖ Before start: submission must be blocked with a message indicating the event has not yet started.
 - ❖ During the event: flag submission must be enabled.
 - ❖ After the end: submission must be blocked, showing a clear message that the event has ended.

- **FR-6.2 – Change event status:** The administrator must be able to change the event status between “not started,” “active,” and “finished” from the administrative interface, with said status being recorded in the database and available to all users within a maximum of 5 seconds after the modification.

FR-6.2 – Change event status (3)**User Story**

As an administrator

I need to change the event status between "not started," "active," and "finished"

So that I can control the event lifecycle and ensure all users have access to the current event state

Acceptance Criteria

- **Given** I am logged in as an administrator and viewing the administrative interface
When I change an event status from "not started" to "active," from "active" to "finished," or to any other valid status
Then the new status is saved in the database and reflected in the administrative interface immediately
- **Given** an administrator has changed an event status
When any user (competitor, captain, or administrator) accesses event information
Then the updated status is visible to them within a maximum of 5 seconds after the modification was made
- **Given** I am logged in as an administrator
When I attempt to change an event status to an invalid value or the system encounters an error during the status change
Then the operation fails, an appropriate error message is displayed, and the previous status remains unchanged in the database

- **FR-6.3 – Reflect event status in the interface:** The system must display the current status of the event ("not started," "active," or "finished") in a visible area of the main interface for all users, updating automatically within a maximum of 5 seconds after any status change.

FR-6.3 – Reflect event status in the interface (3)

User Story

As a User

I need to see the current event status displayed clearly in the main interface

So that I can understand whether the event is not started, active, or finished.

Acceptance Criteria

Given the event has a defined status (“not started,” “active,” or “finished”),

When I am viewing the main interface,

Then the system must display the current event status in a clearly visible area,

And when the event status changes, the displayed status must automatically update to the new value within a maximum of 5 seconds without requiring manual refresh.

- **FR-6.4 – Restrict the sending of flags according to the status of the event:** The system must block the sending of flags to competitors when the status of the event is “not started” or “finished,” and immediately enable it when the status is “active.”

FR-6.4 – Restrict the sending of flags according to the status of the event (5)**User Story**

As a Competitor

I need the system to allow or block flag submissions based on the event status

So that I can only submit flags when the event is active.

Acceptance Criteria

Given the event status is either “not started”, “active”, or “finished”,

When I attempt to submit a flag,

Then the system must block flag submission when the status is “not started” or “finished”,

And the system must allow flag submission immediately when the status is “active”,

And the behavior must reflect the current status at the moment of the attempt.

- **FR-6.5 – Display an informational message when restricting the sending of flags:**
The system must display a non-blocking informational message at the top of the

competitors' interface when flag sending is restricted by the event status (“not started” or “finished”); the message must appear at the time of the sending attempt, remain visible for 5 seconds or until the user performs another action, and then disappear automatically.

FR-6.5 – Display an informational message when restricting the sending of flags (2)**User Story**

As a Competitor,

I need the system to show an informational message when flag submission is blocked due to event status,

So that I understand why I cannot submit a flag.

Acceptance Criteria

Given the event status is “not started” or “finished,”

And I attempt to submit a flag,

When the system blocks the submission due to the event status,

Then the system must display a non-blocking informational message at the top of my interface indicating that flag sending is restricted,

And the message must appear at the moment of the submission attempt,

And the message must remain visible for 5 seconds or until I perform another action, whichever occurs first,

And the message must then automatically disappear.

- **FR-6.6 – Create and edit event rules and guidelines:** The administrator must be able to create and edit the content of event rules and guidelines using a text editor with support for basic Markdown syntax (headings, lists, bold, italics, links, and code blocks) and a maximum limit of 10,000 characters per document.

FR-6.6 – Create and edit event rules and guidelines (8)**User Story**

As an Administrator,

I need to create and edit the event's rules and guidelines using a Markdown-enabled text editor,

So that I can publish and maintain clear and well-structured information for participants.

Acceptance Criteria

Given I am accessing the editor for event rules or guidelines,

When I create or modify content using basic Markdown syntax (headings, lists, bold, italics, links, and code blocks),

Then the system must allow the use of all supported Markdown elements,

And the system must allow saving only when the total content length does not exceed 10,000 characters,

And the editor must enforce the maximum character limit per document.

- **FR-6.7 – Display character counter in the editor:** The system must display a real-time counter within the text editor that indicates the number of characters used out of the 10,000 available.

FR-6.7 – Display character counter in the editor (2)

User Story

As an Administrator,

I need the editor to show a real-time character counter,

So that I can monitor how many characters I have used out of the 10,000 available.

Acceptance Criteria

Given I am viewing or editing the rules or guidelines in the editor,

When I add, modify, or delete text,

Then the system must display an always-visible character counter showing the current number of characters used out of 10,000,

And the counter must update in real time with each change.

- **FR-6.8 – Restrict saving by character limit:** The system must prevent content from being saved when it exceeds the 10,000-character limit and display a non-blocking informational message at the top of the editor indicating that the limit has been exceeded; the message must remain visible for 5 seconds or until the user modifies the text.

FR-6.8 – Restrict saving by character limit (3)**User Story**

As an Administrator,

I need the system to block saving when the character limit is exceeded and notify me,

So that I can adjust the content accordingly before saving.

Acceptance Criteria

Given the editor content exceeds 10,000 characters,

When I attempt to save the content,

Then the system must prevent the save operation,

And the system must display a non-blocking informational message at the top of the editor indicating that the limit has been exceeded,

And the message must remain visible for 5 seconds or until I modify the text, whichever comes first,

And the message must then disappear automatically.

- **FR-6.9 – Render content visible to participants:** The system must render the content of rules and guidelines in a dedicated view accessible to all participants, applying a rendering library that processes basic Markdown and blocks any embedded HTML code or malicious scripts.

FR-6.9 – Render content visible to participants (5)**User Story**

As a System,

I need to render the rules and guidelines in a participant-accessible view,

So that participants can read the content safely with proper Markdown formatting and no

harmful code executed.

Acceptance Criteria

Given rules and guidelines content exists,

When a participant accesses the dedicated public view,

Then the system must render the content using a Markdown rendering library that correctly interprets basic Markdown syntax,

And the system must block any embedded HTML or potentially malicious scripts from being rendered or executed,

And only the sanitized Markdown-rendered content must be visible to participants.

- **FR-6.10 – Reflect changes in the public view:** The system must update the content visible in the participants' public view within a maximum of 5 seconds after an administrator confirms and saves the changes.

FR-6.10 – Reflect changes in the public view (20)

User Story

As a System,

I need to update the public view of the rules and guidelines shortly after changes are saved,

So that participants always see the most up-to-date content.

Acceptance Criteria

Given an administrator has confirmed and saved updated rules or guidelines,

When the save operation completes,

Then the system must update the public participant view to reflect the new content,

And the updated content must become visible within a maximum of 5 seconds without requiring manual page refresh by the participant.

- **FR-6.11 – Define the maximum team size:** The administrator must be able to configure the maximum size allowed per team from the event administration view, storing the value in the database and applying it automatically to team creation and joining validations.

FR-6.11 – Define the maximum team size (2)**User Story**

As an Administrator,

I need to configure the maximum allowed team size,

So that the system enforces this limit during team creation and membership additions.

Acceptance Criteria

Given I am in the event administration view,

When I set or update the maximum team size,

Then the system must store this value in the database,

And the system must apply the configured limit automatically during all team creation and joining validations.

- **FR-6.12 – Restrict the creation of or joining to teams that exceed the limit:** The system must prevent the creation of new teams or the joining of competitors to existing teams when they reach the configured maximum size, displaying a non-blocking information message at the top of the interface for 5 seconds or until the user performs another action.

FR-6.12 – Restrict the creation of or joining to teams that exceed the limit (2)**User Story**

As a Competitor,

I need the system to prevent team creation or joining when the configured size limit is reached,

So that team sizes remain within the allowed restrictions.

Acceptance Criteria

Given the system has a configured maximum team size,

When a competitor attempts to create a new team that would exceed the limit, or attempts to join an existing team already at the limit,

Then the system must block the action,
And the system must display a non-blocking informational message at the top of the interface indicating that the team size limit has been reached,
And the message must remain visible for 5 seconds or until the user performs another action, whichever occurs first,
And the message must then disappear automatically.

- **FR-6.13 – Configure global policy for attempts or flag submission rate:** The administrator must be able to enable and define a global policy that limits the maximum number of attempts or the rate of flag submissions per team or competitor (e.g., maximum of 1 attempt every 10 seconds), which must be stored and automatically applied to all flag submissions.

FR-6.13 – Configure global policy for attempts or flag submission rate (5)

User Story

As an Administrator,

I need to enable and configure a global policy that limits the number or rate of flag submission attempts,

So that the system can automatically enforce this restriction for all teams and competitors.

Acceptance Criteria

Given I am in the administration view for event configuration,

When I enable or modify the global policy defining the maximum allowed attempts or submission rate (e.g., 1 attempt every 10 seconds),

Then the system must store the configured policy in the database,

And the system must automatically apply this policy to all flag submission attempts made by any team or competitor.

- **FR-6.14 – Notify when the flag sending limit is reached:** The system must display a non-blocking informational message at the top of the competitors' interface when the configured attempt limit or rate is reached, indicating the restriction and the time

remaining before the next allowed attempt; the message must remain visible for 5 seconds or until the user performs another action.

FR-6.14 – Notify when the flag sending limit is reached (5)**User Story**

As a Competitor,

I need the system to notify me when I reach the configured attempt or rate limit,

So that I understand why my submission is temporarily blocked and when I can try again.

Acceptance Criteria

Given a global submission limit or rate is configured,

And I attempt to submit a flag but the attempt is blocked due to reaching that limit,

When the system detects the blocked attempt,

Then the system must display a non-blocking informational message at the top of the interface indicating the restriction and the remaining time before another attempt is allowed,

And the message must remain visible for 5 seconds or until I perform another action, whichever occurs first,

And the message must then disappear automatically.

- **FR-6.15 – Apply configuration changes immediately:** The system must immediately apply any changes to global limits (maximum team size or flag submission policy) to all new operations after the modification, without altering historical records or operations already performed.

FR-6.15 – Apply configuration changes immediately (20)**User Story**

As a System,

I need to apply changes to global limits (such as maximum team size or flag submission policy) immediately,

So that all new operations reflect the updated configuration without affecting past records.

Acceptance Criteria

Given an administrator updates a global configuration value (maximum team size or flag submission policy),

When the change is saved,

Then the system must apply the updated configuration immediately to all subsequent operations,

And the system must not modify or retroactively apply changes to any historical records or operations already completed.

FR-7: Maintenance and Deployment

- **FR-7.1 – Provide local deployment using Docker Compose:** The system must provide a local deployment procedure based on Docker Compose that allows frontend, backend, and database services to be launched, ensuring their joint execution in a local environment without requiring cloud dependencies.

FR-7.1 – Provide local deployment using Docker Compose (3)**User Story**

As a System,

I need to provide a local deployment procedure using Docker Compose that launches frontend, backend, and database services,

So that the entire platform can run locally without requiring cloud dependencies.

Acceptance Criteria

Given a local environment with Docker and Docker Compose installed,

When the administrator executes the provided Docker Compose deployment procedure,

Then the system must launch the frontend, backend, and database services together,

And all services must run jointly and functionally without relying on any cloud-based dependency.

- **FR-7.2 – Include orchestration file with minimal configuration:** The system must include a Docker Compose orchestration file that defines the necessary configuration of

services, minimal environment variables, and persistent volumes for the database and challenge files.

FR-7.2 – Include orchestration file with minimal configuration (2)**User Story**

As a System,

I need to include a Docker Compose orchestration file,

So that all required services, minimal environment variables, and persistent volumes for the database and challenge files are properly defined.

Acceptance Criteria

Given the system includes local deployment tooling,

When the administrator inspects the Docker Compose file,

Then the file must define the required services (frontend, backend, database), minimal environment variables needed for startup,

And persistent volumes for both database storage and challenge files must be configured as part of the orchestration.

- **FR-7.3 – Allow deployment with minimal commands:** The administrator must be able to start the entire system environment in a local environment using a maximum of two consecutive commands in the terminal, with the platform accessible from a modern supported browser within a maximum of 60 seconds after execution.

FR-7.3 – Allow deployment with minimal commands (5)**User Story**

As an Administrator,

I need to start the full system locally using no more than two terminal commands,

So that I can quickly run the environment and access the platform in my browser.

Acceptance Criteria

Given Docker Compose is installed and the orchestration file is available,

When I execute at most two consecutive terminal commands to start the environment,
Then the system must launch all required services,
And the full platform must be accessible from a modern supported browser within a maximum of 60 seconds after command execution.

- **FR-7.4 – Retain persistent data between executions:** The system must maintain the data volumes and challenge files configured in Docker Compose persistently between executions, ensuring that re-execution of the deployment does not delete previously stored information.

FR-7.4 – Retain persistent data between executions (8)

User Story

As a System,

I need to maintain persistent data volumes and challenge files between deployments,

So restarting the environment does not delete previously stored information.

Acceptance Criteria

Given the system has been deployed previously and persistent volumes are configured in the Docker Compose file,

When the administrator stops and later re-executes the deployment,

Then all data stored in the configured database volumes and challenge file volumes must remain intact,

And the re-execution must not delete or overwrite previously stored information.

- **FR-7.5 – Maintain persistent volumes for database and challenge files:** The system must maintain separate persistent volumes for the database and for files associated with challenges, ensuring that information is preserved between environment restarts and local deployment updates.

FR-7.5 – Maintain persistent volumes for database and challenge files (5)

User Story

As a System,

I need to maintain separate persistent volumes for the database and challenge files,
So that their data is preserved across environment restarts and deployment updates.

Acceptance Criteria

Given persistent volumes for the database and challenge files are configured,

When the environment is restarted or updated locally,

Then the system must preserve all data stored in each respective volume,

And the database data must remain intact in its dedicated volume,

And challenge file data must remain intact in its separate dedicated volume,

****Without requiring manual administrator intervention.**

- **FR-7.6 – Define separate volumes in deployment orchestration:** The Docker Compose orchestration file must explicitly define two separate persistent volumes (one for the database and one for challenge files), specifying their storage paths in the deployment configuration.

FR-7.6 – Define separate volumes in deployment orchestration (3)**User Story**

As a System,

I need to define two separate persistent volumes in the Docker Compose file,
So that the database and challenge files have distinct storage paths configured at deployment.

Acceptance Criteria

Given Docker Compose is used to orchestrate local deployment,

When the administrator inspects the orchestration file,

Then the file must explicitly define one persistent volume for database storage and another for challenge file storage,

And each volume must include its own storage path within the deployment configuration.

- **FR-7.7 – Preserve data after service restart:** The system must ensure that when any of the environment's services (frontend, backend, or database) are restarted, the data stored in the persistent volumes remains intact and accessible, without requiring manual intervention by the administrator.

FR-7.7 – Preserve data after service restart (5)**User Story**

As a System,

I need to keep all data stored in the persistent volumes intact when any service is restarted,
So that normal service restarts do not cause data loss or require manual recovery.

Acceptance Criteria

Given the environment's services (frontend, backend, or database) are running with their configured persistent volumes,

When any individual service is restarted,

Then the data stored in the persistent volumes must remain intact and fully accessible to the service once it resumes,

And no manual administrator action must be required to restore or reattach the data.

- **FR-7.8 – Allow volumes to be relocated through configuration:** The administrator must be able to modify the paths of persistent volumes using documented configuration parameters, maintaining the integrity of existing data during the relocation.

FR-7.8 – Allow volumes to be relocated through configuration (20)**User Story**

As an Administrator,

I need to modify the paths of persistent volumes through documented configuration parameters,

So that I can relocate volumes while preserving existing stored data.

Acceptance Criteria

Given the system provides documented configuration parameters for persistent volume paths,
When I update the configuration to relocate either the database volume or the challenge files volume,
Then the system must apply the new paths while preserving all existing data within those volumes,
And the relocated volumes must remain fully accessible after the configuration change.

- **FR-7.9 – Update the environment without losing persistent data:** The administrator must be able to update the local system environment—including container images and minimum environment variables—without deleting or modifying the information stored in persistent database volumes and challenge files.

FR-7.9 – Update the environment without losing persistent data (5)

User Story

As an Administrator,

I need to update the local environment (container images and minimum environment variables) without losing existing persistent data,

So that stored information in database and challenge volumes remains intact after updates.

Acceptance Criteria

Given the system uses persistent volumes for database and challenge files,

When I update the environment by replacing container images and adjusting minimal environment variables,

Then the system must preserve all data stored in those persistent volumes,

And no stored information may be deleted, modified, or corrupted as a result of the update process.

- **FR-7.10 – Provide a documented update procedure:** The system must include a documented procedure indicating the steps necessary to stop services, update images, and restart the environment, ensuring the integrity of data and files during the process.

FR-7.10 – Provide a documented update procedure (13)**User Story**

As a System,

I need to provide a documented procedure describing the steps to stop services, update images, and restart the environment,

So that administrators can perform updates while preserving system integrity.

Acceptance Criteria

Given an administrator needs to update the environment,

When they consult the system's update documentation,

Then the documentation must clearly describe the steps to stop services, update container images, and restart the environment,

And it must explicitly ensure that following the procedure preserves data and challenge files stored in persistent volumes.

- **FR-7.11 – Restart environment after update while preserving configuration:** The system must start correctly after an update within a maximum of 60 seconds, preserving the configuration, scores, and statuses of previous challenges, without generating inconsistencies or loss of information.

FR-7.11 – Restart environment after update while preserving configuration (5)**User Story**

As a System,

I need to restart correctly after an update within 60 seconds,

So that all previous configuration, scores, and challenge statuses remain intact and consistent.

Acceptance Criteria

Given the environment has been updated following the documented procedure,

When the system is restarted,

Then it must become fully operational within a maximum of 60 seconds,

And all previous configuration values, scores, and challenge statuses must be preserved exactly as they were before the update,

And no inconsistencies or information loss may occur as a result of the restart.

- **FR-7.12 – Manage schema migrations in a controlled manner:** The system must detect when an update requires a database schema migration and explicitly indicate this requirement in the update procedure; the migration must be executed atomically and securely or, otherwise, blocked by displaying a blocking information message to the administrator with the cause of the block.

FR-7.12 – Manage schema migrations in a controlled manner (20)

User Story

As a System,

I need to detect when a database schema migration is required and handle it securely,

So that schema changes occur atomically or are blocked with an explicit explanation to the administrator.

Acceptance Criteria

Given an update is being performed,

And the system detects that a schema migration is required,

When the administrator consults or executes the update procedure,

Then the need for the migration must be explicitly indicated in the documented procedure,

And the migration must execute atomically and securely,

Or, if atomic execution is not possible, the system must block the migration and display a blocking informational message specifying the reason for the block.

- **FR-7.13 – View basic service status:** The administrator must be able to view, from the administrative interface, the basic status of each main system service (frontend, backend, and database), showing whether they are “running” or “stopped,” with an automatic update every 10 seconds.

FR-7.13 – View basic service status (5)**User Story**

As an Administrator,

I need to view the basic status of each main system service (frontend, backend, database),

So that I can confirm whether services are running or stopped from the administrative interface.

Acceptance Criteria

Given the administrative interface includes a service status section,

When I access this section,

Then the system must display for each main service (frontend, backend, database) whether it is “running” or “stopped,”

And the displayed status must automatically update every 10 seconds without requiring manual refresh.

- **FR-7.14 – View sensitive operational logs:** The administrator must be able to access recent operational logs for each service from the administrative interface, with automatic pagination or truncation that limits the query to the last 1,000 records per service, without affecting overall system performance.

FR-7.14 – View sensitive operational logs (5)**User Story**

As an Administrator,

I need to access recent operational logs for each service,

So that I can review the latest 1,000 records without affecting system performance.

Acceptance Criteria

Given the administrative interface provides access to service logs,

When I request logs for a specific service,

Then the system must return only the most recent records, limited to a maximum of 1,000

entries through pagination or truncation,

And the retrieval of these logs must not degrade overall system performance.

- **FR-7.15 – Ensure non-intrusive status and log queries:** The system must ensure that service status and administrative log query operations do not block or degrade normal user operations or environment performance.

FR-7.15 – Ensure non-intrusive status and log queries (5)

User Story

As a System,

I need to ensure that service status checks and log queries do not interfere with normal platform operations,

So that users can continue using the environment without performance degradation.

Acceptance Criteria

Given administrators may request service statuses or recent logs,

When these operations are executed,

Then they must not block or slow down normal system operations for competitors, captains, or other users,

And overall environment performance must remain unaffected while status or log queries are processed.